Acids Investigation

Aim: To find out the properties of acids.

Write down what you already know about acids.

_____
_____
_____
_____
_____

Write down the names of any acids that you know of.

_____
_____

Problem: You have a range of chemicals called acids. Your task is to perform a series of tests to investigate the properties of acids. At the end of the investigation you should make a list of the properties of acids.

Equipment: A range of common, dilute acids in dropper bottles
Red and blue litmus paper
Pieces of magnesium ribbon
Samples of different metals
Dilute hydrochloric acid
Test tubes with rubber stoppers

List any safety procedures you should follow:

_____
_____
_____
_____
_____

Procedure:
Part 1 – Litmus Test
Place a couple of drops of each acid on separate pieces of red and blue litmus paper. Record your observations in the table below.

Name of Acid	Red Litmus Paper	Blue Litmus Paper

Acids will

_____

_____

Part 2 – Magnesium Ribbon Test
Place a small piece of magnesium in a test tube.
Chose an acid from test one and add enough to cover the magnesium.
Record your observations in the table below.
Do this for two more acids.

Name of Acid	Observations

Acids will

_____

_____

Part 3 – Acids and Metals
Place a small sample of each metal in a test tube.
Add enough dilute hydrochloric acid to each test tube to cover the metal.
Record your observations in the table below.

Name of Metal	Observations

Acids will

_____

_____

Part 4

Think of foods you eat which can be classified as acids. List them below.

_____

_____

Describe what these foods taste like.

_____

_____

Conclusions:

From your observations, list the common properties of acids.

_____

_____

_____

_____

_____

Complete the table below.

Name of Acid	Chemical Formula	Number of Atoms of Each Element
	HCl	
Nitric Acid		1 Hydrogen 1 Nitrogen 3 Oxygen
Sulfuric Acid	H_2SO_4	
Phosphoric Acid		3 Hydrogen 1 Phosphorus 4 Oxygen

What element is common to all of these acids?

_____

Why must you be careful when working with acids?

_____
_____
_____
_____